Effect of wheat bran on serum lipids: influence of particle size and wheat protein.

OBJECTIVE: Wheat fiber appears to protect from cardiovascular disease despite its lack of consistent effect on serum lipids. We therefore wished to determine whether reported inconsistencies in the effect of wheat bran resulted from differences in particle size or its high gluten content. METHODS: Two studies were conducted. In one-month metabolic diets, 24 hyperlipidemic subjects consumed breads providing an additional 19 g/d dietary fiber as medium or ultra-fine wheat bran and extra protein (10% of energy as wheat gluten). In two-week ad libitum diets, 24 predominantly normolipidemic subjects consumed breakfast cereals providing an additional 19 g/d of dietary fiber as coarse or a mixture of ultra-fine and coarse wheat bran with no change in gluten intake. Both studies followed a randomized crossover design with control periods when subjects ate low-fiber breads and cereals respectively with no added gluten. Fasting blood lipids were measured on day zero and at the end of each phase. RESULTS: Wheat bran had no effect on total, LDL or HDL cholesterol irrespective of particle size or level of gluten in the diet. However, consumption of increased gluten in the metabolic study was associated with a 13+/-4% reduction in serum triglycerides ($p = 0.005$) which was not seen in the normal-gluten ad libitum study. CONCLUSIONS: The protective effect of wheat fiber in cardiovascular disease cannot be explained by an effect of wheat bran in reducing serum cholesterol although in hyperlipidemic subjects displacement of carbohydrate by gluten on the high-fiber phases was associated with lower serum triglycerides.